

Compass IoT: Driving a safer future on Australian roads with Google Cloud

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ABOUT COMPASS IOT

Compass IoT launched its groundbreaking road intelligence platform, enabling the volume of data it monitors to double each year while leveraging Google Cloud to make communities safer after major floods and a global expansion with Google Cloud.



About Compass IoT

Compass IoT is an award-winning Australian road Intelligence company that uses connected vehicle data to improve road safety, infrastructure and city planning. Since 2018, Compass IoT has set out to make transport data more accessible and simple to understand, with solutions that are easy to use for entities ranging from municipal governments to highway operators. Cloud-based IoT innovation has enabled Compass IoT to be named 'Best Data-driven Startup' at the 2019 Australasian Start-up Awards, and 'Data Strategy Partner' with the

Google Cloud results

- Supports yearly doubling of connected vehicle network with unlimited data processing power
- Enables mission of real-time road monitoring with data pipelines of latencies under 100 milliseconds
- Justifies \$73 million bridge project by identifying harsh braking trend with BigQuery enabled

Scales
than 1
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vehicl

Government of the Australian Capital Territory in 2020.

Industries: Technology

Location: Australia

SQL (<https://cloud.google.com/sql/>),

Google Cloud (<https://cloud.google.com/>)
(<https://cloud.google.com/kubernetes-engine/>),

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://cloud.google.com/pubsub/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

data
intelligence

- Supports
global
growth
strategy
with data
center
network
enabling
minimum
latency
connectivity
and GDPR
compliance

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That's made
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—Emily Bobis, Co-founder,
Compass IoT

When massive flooding hit Lismore, a city in Australia's New South Wales, the damage (<https://www.theguardian.com/australia-news/2022/apr/02/adrift-in-a-parallel-universe-lismore-struggles-to-find-itself-after-another-flood>) affected more than just homes and businesses. The corrosive tide that included sewage and toxic waste also eroded the quality of Lismore's roads, creating a hidden hazard for the community.

Compass IoT (<https://www.compassiot.com.au/>), a startup that uses connected vehicle data to improve road safety and urban planning, crunched data on factors such as swerving and roll pitch from thousands of cars to establish that the flooding caused a 50% drop in road quality. The intelligence prompted local authorities to prioritize repaving of Lismore's roads, directly benefiting community welfare.



The Lismore case illustrates how Compass IoT is driving a new paradigm of Internet of Things (IoT) enabled road safety, winning it the "best data-driven startup award" at the Australasian startup awards. The platform ingests millions of data points from connected vehicles in real-time, on factors including speed, acceleration, near misses and traction, and runs them through machine learning algorithms that yield granular road intelligence.

Clients from city planners to tunnel operators can access the data insights they need, anywhere in Australia, on a convenient and intuitive dashboard, for use cases such as where to build a new level crossing or how to improve safety with a new traffic light.

To power its mission, Compass IoT needed powerful data processing and analytics tools available, as well as the agile autoscaling of an infrastructure built on microservices. By turning to Google Cloud (<https://cloud.google.com/>) and a combination of Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>)(GKE), BigQuery (<https://cloud.google.com/bigquery>) and

Pub/Sub (<https://cloud.google.com/pubsub>), the startup unlocked the optimized data solutions it needed to transform road safety in Australia.

"To study road safety in the past, you needed to set up cameras and obtain location licenses. Our IoT solutions track Australia's entire road network in real-time, with no hardware deployments," says Emily Bobis, Co-founder, Compass IoT. "That's made possible by the autoscaling and unmatched data processing offered by Google Cloud."

"The importance of the speed of data pipelines enabled by Pub/Sub cannot be overstated. If you're trying to run a real-time network, and see a crash on the network, a minute makes a big difference in a situation

where lives are
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—Angus McDonald, Co-
founder, Compass IoT

Enabling exponential growth in vehicle monitoring on data superhighways

A typical dashboard visualization of the Compass IoT solution shows a river of green and orange dots that reconfigures second-by-second. The dots represent cars on a road, with green signaling normal driving and orange an aberration such as swerving, speeding or harsh braking. The heatmap gives a granular real-time snapshot of road conditions anywhere in Australia.

Aggregated over time, clients can use these data insights to prioritize infrastructure building or repairs, depending on spots in the network that show the most strain. The platform has achieved successes ranging from establishing that a new tunnel halved the number of accidents on a busy Sydney road, to justifying a \$73 million bridge-widening project by identifying a trend in harsh braking.

On a normal day, Compass IoT monitors roughly 700,000 vehicles on Australia's roads, yet that can scale up to more than 1.5 million at any time. The company relies on GKE for the seamless and flexible

computing power it needs to scale up-and-down according to highly unpredictable traffic volume. It is increasingly turning to Cloud Run (<https://cloud.google.com/run/>) to obtain even more agile and affordable compute power in a serverless environment.

According to Angus McDonald, Co-founder, Compass IoT, Pub/Sub plays a key role as the platform's data ingestion tool by enabling unlimited data pipelines with latencies under 100 milliseconds. The data can then be fed in real-time into BigQuery to churn out the data intelligence clients rely on to improve safety on Australia's roads.

"Google Cloud
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analytics at any

given moment,
and allow us to
comply with
European data
privacy
regulations."

*—Angus McDonald, Co-
founder, Compass IoT*

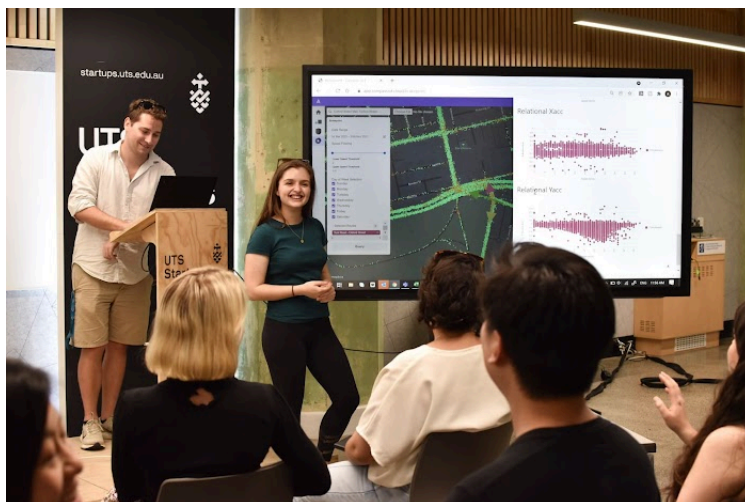
Pub/Sub pulls in data packets from each vehicle and ingests it into Cloud SQL (<https://cloud.google.com/sql>), the platform's managed database, with time-stamps at both ends showing a minimum time of 5 seconds and a maximum time of 46, says McDonald. This enables the real-time monitoring of road conditions that Compass IoT needs to provide value in its mission.

"The importance of the speed of data pipelines enabled by Pub/Sub cannot be overstated," says McDonald. "If you're trying to run a real-time network, and see a crash on the network, a minute makes a big difference in a situation where lives are at stake."

"The fact that it's coming through the whole system in under 46 seconds, in the worst case scenario, and under five seconds, in the best case scenario," he adds, "is a huge enabler in our goal mission to keep Australian drivers safe."

Overall, McDonald says, the unlimited compute and data processing power enabled by a combination of Pub/Sub, BigQuery, GKE and Cloud Run has allowed Compass IoT to double the number of cars the platform monitors every year, from 200,000 in 2018 to today's volume of more than 1.5 million.

Taking optimized transport intelligence to global roads with Cloud CDN



Compass IoT now plans to bring its road intelligence software to a global market, speaking with transport authorities in the United Kingdom, Austria and Greece about deploying the platform on their networks. To enable this goal, the startup says it will depend heavily on Cloud CDN

(<https://cloud.google.com/cdn/>) for seamless, automated connections to Google Cloud's global network of data centers, for the lowest latency content delivery possible.

Google Cloud's world class data center network will also enable Compass IoT to operate in Europe by complying with General Data Protection Regulations (GDPR) that require personal data such as those that come from private vehicles to be stored within Europe.

"Google Cloud has seamless global reach with its unrivaled network of data centers," says McDonald.

"This is going to enable our global expansion by providing the lowest latency for our real-time analytics, and allow us to comply with European data privacy regulations."